

Stability-Aware and Risk Management Frameworks for SFC Allocation and Association in LEO Satellite Networks

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Talk Roadmap

- 1 Introduction and Research Problems
- 2 Visibility-Aware User Association and Resource Allocation
- 3 Stability-Aware Security Function Chain (SFC) Orchestration and Associations
- 4 Risk-aware SFC Allocation
- 5 Conclusion
- 6 Questions

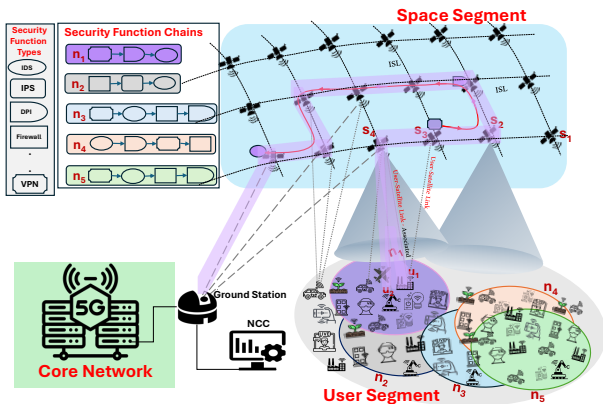
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System Model

- 1 Users request slice-specific services.
- 2 Network Slicing.
- 3 Security Function Chains (SFCs).
- 4 Each slice has an ordered SFC.
- 5 LEO satellites provide relay & compute; ISLs connect satellites.
- 6 Ground stations provide core connectivity.



Research Problems

First Problem

How to fairly and stably associate users with satellites, while meeting per-slice quality-of-service requirements.

Second Problem

How to tackle the joint problem of user–satellite association, satellite–ground station association, and security functions allocation while minimizing disruptive reconfigurations.

Third Problem

How to allocate and share security functions across slices (i.e. applications) while balancing risk and resource use.

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Visibility-Aware User Association and Resource Allocation

Proposed Solution

- Optimizing the demand satisfaction ratio (DSR)
 - Maximizing the minimum DSR to ensure that the data requirements of each user are met.
- Delay optimization
 - Minimizing the end-to-end propagation delay for all users
- Stability management
 - Maximizing the stability of user-satellite connections to reduce the frequency of handovers.

IEEE Transactions on Network and Service Management, under review

Visibility-Aware User Association and Resource Allocation in Multi-Slice LEO Satellite Networks

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Abstract—The low Earth orbit (LEO) satellite megaconstellation can provide ubiquitous coverage and high-performance connectivity, supporting multi-slice applications with various key performance indicator (KPI) requirements. However, due to the dynamic nature of LEO satellites, limited resources, and the diverse demands of different slices, managing user association (UA) and resource allocation becomes an increasingly challenging task in areas with overlapping satellite coverage. This paper proposes a joint optimization model for UA and resource allocation in multi-slice LEO satellite networks (SLSNs). Based on mixed-integer non-linear programming (MILP), our model minimizes the total propagation delay and optimizes the demand satisfaction ratio (DSR) using a Max-Min approach to ensure each slice meets its unique throughput requirements. In addition, a visibility-aware component is incorporated to prioritize longer satellite visibility, reduce handovers, and improve network stability. Due to the computational complexity of the MILP model, we propose a heuristic-based balanced association with delay-aware bandwidth

progress has been made in deploying these constellations, driven by initiatives from SpaceX, OneWeb, Amazon, and Huawei. For instance, SpaceX's Starlink project has deployed 6,290 operational satellites¹ with plans for further expansion to enhance global coverage. These advancements ensure that more than 30 satellites are within range of most populated areas, providing unprecedented connectivity. Similarly, OneWeb's constellation, the second largest operational LEO network [3], and Amazon's Project Kuiper aim to deliver high-speed, low-latency access critical for supporting emerging technologies [4].

Network slicing (NS) is a pivotal technology for enabling the adaptability and flexibility required to meet the diverse demands of next-generation networks [5], [6]. By creating isolated virtual networks over shared infrastructure, NS allows operators to tailor resources to specific service

Visibility-Aware User Association and Resource Allocation

Problem Formulation: MINLP-based Optimization Approach

- Objective: Jointly optimizing user-satellite associations and bandwidth allocation to balance three objectives:

$$\max_{\alpha, w} \left\{ \beta_d \cdot D - \beta_p \cdot \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}_n} \sum_{s \in \mathcal{S}} T_{u,n,s}^{\text{prop}} \cdot \alpha_{u,n}^s + \beta_v \cdot \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}_n} \sum_{s \in \mathcal{S}} \sum_{g \in \mathcal{G}} (\nu_{n,u}^s + \nu_g^s) \cdot \alpha_{u,n}^s \right\}, \quad (1)$$

where,

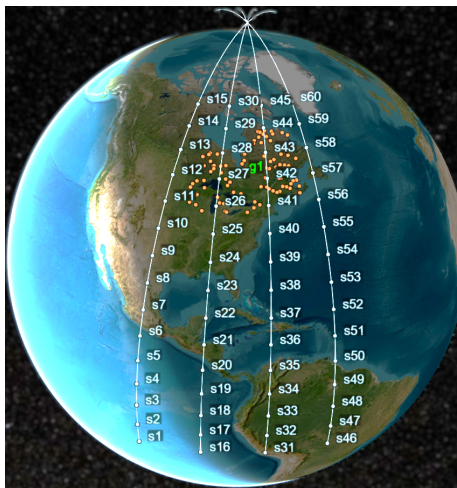
$$D = \min_{n \in \mathcal{N}, u \in \mathcal{U}_n} \text{DSR}_{u,n}, \quad (2)$$

- To avoid under-utilization when maximizing fairness, a bisection algorithm is used to find the optimal minimum DSR threshold.
- To overcome the high complexity of the MINLP, Balanced association with Delay-Aware bandwidth Distribution (B-DAD) is introduced.

Visibility-Aware User Association and Resource Allocation

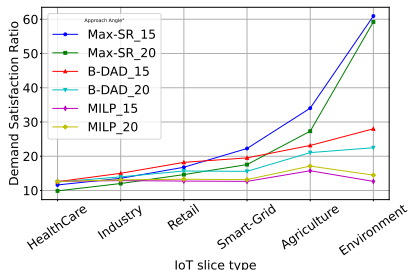
Simulation Setup: Satellite Scenario

- Matlab SatCom Toolbox is used to simulate the satellite scenarios
- Gurobi solver

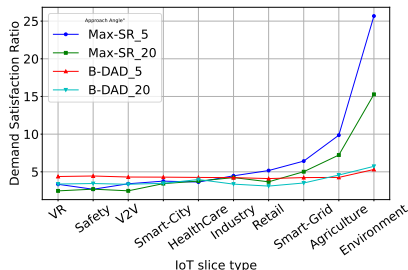


Visibility-Aware User Association and Resource Allocation

Selected Results: Demand Satisfaction Ratio (SDR)



(a) Small-scale test case

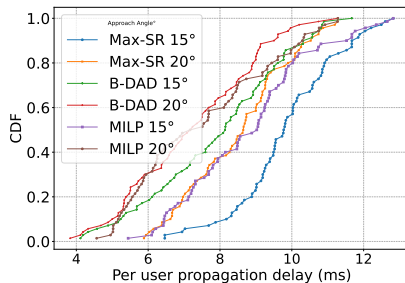


(b) Large-scale test case

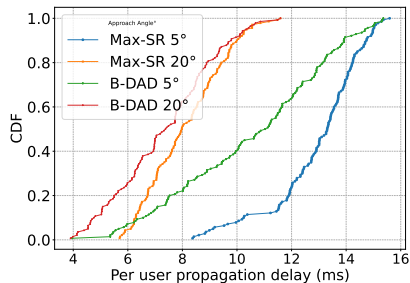
Figure: Comparison of average DSR of slices for different approaches and elevation angles

Visibility-Aware User Association and Resource Allocation

Selected Results: CDF of Propagation Delay



(a) Small-scale test case

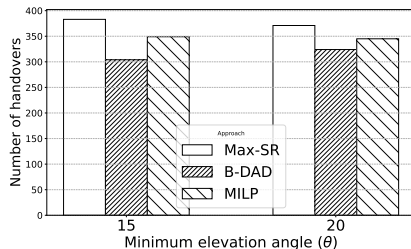


(b) Large-scale test case

Figure: Comparison of propagation delay for different approaches and elevation angles

Visibility-Aware User Association and Resource Allocation

Selected Results: Network Stability [Handovers]



(a) Small-scale test case

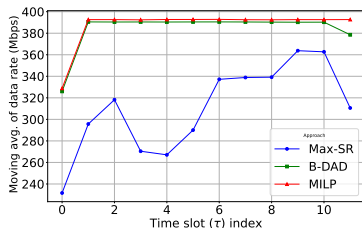


(b) Large-scale test case

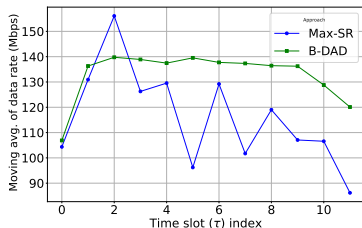
Figure: Comparison of the number of handovers for different approaches and elevation angles

Visibility-Aware User Association and Resource Allocation

Selected Results: Network Stability [Data Rate Smoothness]



(a) Small-scale test case

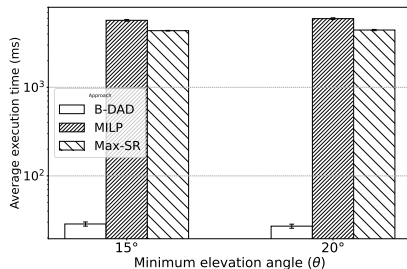


(b) Large-scale test case

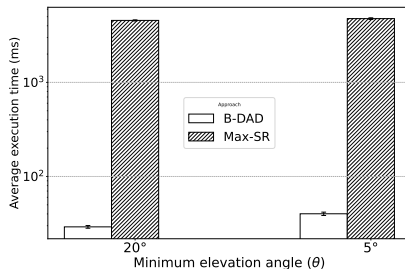
Figure: Data rate moving averages for healthcare slice with 20° elevation angle and network scales.

Visibility-Aware User Association and Resource Allocation

Selected Results: Execution Time



(a) Small-scale test case



(b) Large-scale test case

Figure: Comparison of execution time for different approaches and elevation angles

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Stability-Aware SFC Orchestration and Associations

Motivations and Problem Statement

- **LEO networks are highly dynamic:** satellite visibility 4–10 minutes $\Rightarrow$ many handovers and topology changes.
- **Conventional Approach Limitations:** Traditional per-window (time-step) optimizations recompute SFC placement and routing from scratch each interval.
- **Conventional Solutions are Fragmented:** Past works often optimized user association *or* SFC placement in isolation.
- **Scaling Challenges:** As constellations and user demands grow, continuous re-optimization becomes unsustainable..
- **Stability as a Priority:** Ensuring operational continuity (minimal disruptions) is essential for SLA compliance in LEO networks.
- **Gap in Literature:** No existing solution provides a *stability-aware, joint optimization* of SFC orchestration and user–satellite–GS association in LEO networks. This gap motivates the STARS framework.



Stability-Aware SFC Orchestration and Associations Outcomes

IEEE Wireless Communications and Networking Conference (WCNC), 2026, under review

Joint Satellite Association and SFC Placement with Stability Optimization in LEO Satellite Networks

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Abstract—Low Earth orbit (LEO) mega-constellations enable global, low-latency connectivity but challenge the security function chain (SFC) orchestration due to fast-changing visibility, resource volatility, and heterogeneous quality of service (QoS) requirements. To tackle this issue, we present in this paper SATStab, a stability-aware orchestration framework that co-designs: (i) stability-regulated objectives penalizing configuration churn across time windows; (ii) temporal decoupling with visibility-aware repair and warm start; and (iii) hierarchical decoupling that separates fast association decisions from slower SFC placement and resource allocation. We formulate SATStab as a mixed-integer non-linear programming (MINLP) problem and evaluate it with real-time LEO dynamics. We solve it through a two-stage approach, where in the first stage, we optimize satellite-user associations while in the second stage, we optimize SFC placement. Through extensive simulations, we show that, relative to a monolithic MINLP, SATStab reduces user-satellite handovers by 54.3%, satellite-ground station (GS) reassociations by 59.3%, and SFC migrations by 32.7%, while cutting solution time by 56.8% without degrading end-to-end (E2E) delay. These results prove the efficacy of SATStab in terms of resource orchestration and resilience.

Index Terms—GG, LEO, Satellite, Security Function Chain, Network Slicing, Association, MINLP, Optimization, Allocation.

I. INTRODUCTION

The recent launch and rapid scaling of Low Earth orbit (LEO) satellite mega-constellations such as SpaceX Starlink and OneWeb are reshaping global connectivity by providing low-latency broadband to remote, underserved areas and mobile users [1]. These constellations combine wide-area coverage with increasingly capable on-board computing and inter-satellite links (ISLs), enabling novel paradigms such as orbital edge computing and in-space network function virtualization (NFV) [2]. At the same time, emerging 6G visions emphasize network softwareification, flexible network slicing (NS), and E2E service orchestration to satisfy diverse vertical requirements from ultra-reliable and low latency communications (URLLC) to high-throughput, for industrial control, security-sensitive applications, and emergency response [3]. Enabled by NFV and software-defined network (SDN), NS allows operators to instantiate virtual slices with tailored quality of service (QoS) and security policies over a shared physical infrastructure [4]. For security-sensitive slices, reduced chains of virtualized security functions (SFs) such as firewalls, intrusion detection system (IDS), and

encryption modules, virtual private network (VPN) are deployed as security function chain (SFC) to meet regulatory, confidentiality, and integrity requirements [5].

In terrestrial clouds and edge data centers, SFC orchestration and virtual network function (VNF) placement have been extensively studied, yielding methods that trade off placement cost, delay, reliability, and resource utilization [6]. However, when moving such capabilities into the LEO satellites' domain, new challenges emerge due to the unique spatiotemporal properties of satellite networks. LEO satellites traverse a given GS location in a few minutes, generating rapidly evolving visibility windows for user terminals and GSs, frequent handovers, and time-varying ISL topologies [7]. These dynamics fragment available processing resources and alter feasible routing paths toward GS gateways on short timescales, undermining the continuity assumptions of traditional per-time window optimization approaches. Consequently, sequential re-optimization at each scheduling window, common in many prior terrestrial and non-terrestrial studies [8]–[15], can cause excessive reconfigurations, e.g., frequent user-satellite handovers, satellite-GS re-associations, and repeated SFC redeployments or SF migrations. Such changes increase signaling overhead, incur state transfer and migration costs, particularly harmful for stateful SFs, and can degrade the user experience via transient packet loss and delay spikes [16].

Existing literature has made important progress in (i) user association and resource allocation for LEO and hybrid networks [11], [12], [17], [18], and (ii) SFC/VNF placement and routing using ILP, heuristics, and machine learning [6], [14], [15], [19]. However, these strands are often treated independently, i.e., user association and radio/resource management are optimized without jointly considering SFC placement and end-to-end (E2E) routing to GSs, while SFC algorithms typically assume static or slowly changing connectivity and seldom penalize frequent reconfigurations across consecutive time windows. Moreover, some works abstract the satellite-GS access path, ignoring a critical component of the E2E path that can substantially affect delay and feasibility when SFCs are placed in the space segment.

The confluence of these gaps motivates new orchestration

IEEE Transactions on Mobile Computing, under review

STARS: Stability-Aware SFC Orchestration and Associations in LEO Satellite Networks

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Abstract—Low Earth orbit (LEO) satellite networks present critical challenges for security function chains (SFC) orchestration and associations due to rapid topology changes, resource volatility, and heterogeneous service requirements that render conventional SFC optimization approaches ineffective. To tackle this issue, we introduce here STARS, an optimization framework that fundamentally transforms the sequential time-window optimization for SFC orchestration and satellite association through three techniques: (1) Stability-aware regularization that penalizes configuration changes across time windows, thus reducing handovers by 54% and security function migrations by 33%; (2) Temporal decoupling that leverages solutions from prior time windows as warm-start seeds and dynamic repairing using real-time visibility constraints; and (3) Hierarchical decoupling that separates satellite association and SFC placement into computationally efficient stages, thus reducing time complexity. Through rigorous formulation as a mixed-integer non-linear programming (MINLP) and simulation-based evaluation, STARS achieves a 57% reduction in optimization solution time, a 7% reduction in the load of deployed security function instances, and efficient CPU utilization (9.81% increase) compared to benchmark schemes. STARS delivers these substantial benefits without any degradation in end-to-end delay. Note that the reported performance values are based on our specific system parameter choices and simulation setup and may not be universally representative. The co-design of stability mechanisms and decoupling strategies establishes STARS as a new paradigm for resilient satellite network optimization, balancing optimality, continuity, and computational tractability under high LEO satellite dynamism.

Index Terms—Security Function Chain, user association, LEO satellite, 6G, network slicing, stability.

I. INTRODUCTION

THE deployment of LEO satellite mega-constellations, such as SpaceX Starlink and OneWeb, transforms global connectivity by extending reliable network access to remote and underserved regions [1]. For example, Starlink has launched more than 7,800 operational satellites¹, enabling broadband access in rural areas, disaster zones, and for emergency responders where terrestrial infrastructure is absent or damaged [2], [3]. The growing processing capabilities of onboard satellites allow these platforms to function as distributed computing nodes, enabling intelligent edge services [4], such as smart agriculture, healthcare, public safety, virtual reality (VR), and vehicle-to-vehicle (V2V) communications [5], which require low-latency and

The network slicing (NS) paradigm, enabled by network function virtualization (NFV) and software-defined network (SDN), allows the creation of multiple virtual networks with distinct quality of service (QoS) and security requirements [7] over shared physical infrastructure [8], [9]. Recognized by the 3rd generation partnership project (3GPP) as critical for 6G virtualization [10], [11], NS supports diverse application needs, ranging from real-time remote surgery to industrial automation [6]. In satellite networks, NS plays a pivotal role by enabling efficient resource allocation and services' isolation in highly dynamic environments, while ensuring QoS requirements. However, the broadcast nature of satellite communications introduces cyber-physical security risks, such as jamming, spoofing, and eavesdropping [12], thus necessitating robust countermeasures. To address these vulnerabilities, security functions (SFs) such as firewalls, intrusion detection system (IDS), intrusion prevention system (IPS), virtual private network (VPN), encryption modules, and authentication gateways can be integrated into network slices to protect both user data and control traffic [13]. SFs are critical for safeguarding services and preserving the overarching integrity, confidentiality, and availability of the satellite network infrastructure [14].

The security function chains (SFCs) provide the framework to systematically enforce protections through ordered/chaired SFs (e.g., firewall → IDS → encryption), ensuring compliance with service-level agreements (SLAs) and regulatory needs for critical communications [15], [16]. For instance, a typical SFC might involve packet inspection by a firewall, anomaly detection by an IDS, followed by encryption and secure authentication, each tailored to the specific security service context [17]. Furthermore, as satellite networks increasingly leverage NFV and dynamic resource allocation [18], SFCs allow for scalable, programmable, and orchestrated deployment of security services, thus rapidly adapting to evolving threat landscape and dynamic satellite networks [19]. Although SFCs are able to mitigate threats, their effectiveness depends on stable orchestration across dynamic topologies [15]. Specifically, LEO orbital dynamics complicate SFC orchestration as satellites traverse user coverage zones in brief 4–10 minute time windows, resulting in rapid changes in visibility, frequent handovers, and fragmented processing

Stability-Aware SFC Orchestration and Associations

Decision Variables (per window)

- $\alpha_{u,n}^s$: **user–satellite** association.
- $\xi_{u,n}^{s,g}$: **satellite–GS** routing association.
- $\beta_{u,n,f}^{i,s}$: **SFC placement** (instance i of function f on satellite s).

Constraints: visibility, capacity, flow/order, processing resources, throughput, and end-to-end delay.

Stability-Aware SFC Orchestration and Associations

End-to-End Delay Components

- **Propagation:** user→sat, ISLs between SF hosts, last-SF→routing sat, sat→GS.
- **Processing:** per-function load vs. allotted CPU (MIPS).
- **Transmission:** access rate dependent on bandwidth share & SINR.

QoS constraint

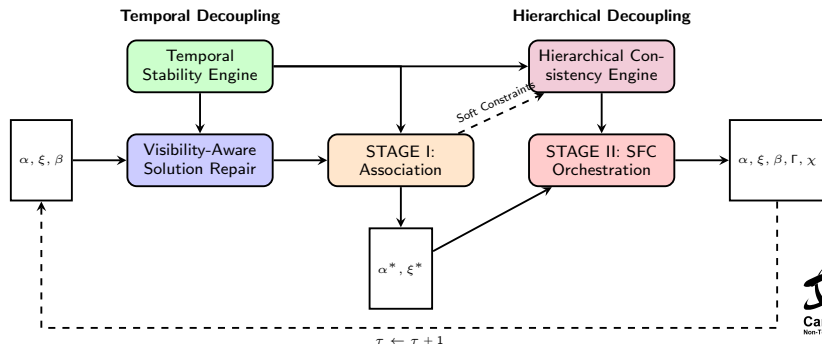
$$\delta_{n,u}^{\text{tot}} = \delta_{n,u}^{\text{prop}} + \delta_{n,u}^{\text{proc}} + \delta_{n,u}^{\text{tx}} \leq D_n$$

$$\begin{aligned} \delta_{n,u}^{\text{prop}} = & \underbrace{\sum_{s \in \mathcal{S}} \alpha_{u,n}^s \cdot d_{u,n}^s}_{\text{Term 1: User to Satellite}} + \underbrace{\sum_{s \in \mathcal{S}} \alpha_{u,n}^s \left(\sum_{\dot{s} \in \mathcal{S}} \sum_{i \in \mathcal{I}_{\mathcal{F}_n\{1\}}} \beta_{u,n,\mathcal{F}_n\{1\}}^{i,\dot{s}} \cdot d_s^{\dot{s}} \right)}_{\text{Term 2: User-Satellite to 1st SF}} + \underbrace{\sum_{p \in \mathcal{P}_n} \sum_{s \in \mathcal{S}} \sum_{s' \in \mathcal{V}_s} \chi_{n,u,p}^{s,s'} \cdot d_s^{s'}}_{\text{Term 3: Inter-SF Routing}} \\ & + \underbrace{\sum_{s \in \mathcal{S}} \sum_{g \in \mathcal{G}} \xi_{u,n}^{s,g} \left(\sum_{\dot{s} \in \mathcal{S}} \sum_{i \in \mathcal{I}_{\mathcal{F}_n\{F\}}} \beta_{u,n,\mathcal{F}_n\{F\}}^{i,\dot{s}} \cdot d_s^{\dot{s}} \right)}_{\text{Term 4: Last SF to Routing Satellite}} + \underbrace{\sum_{s \in \mathcal{S}} \sum_{g \in \mathcal{G}} \xi_{u,n}^{s,g} \cdot d_s^g}_{\text{Term 5: Satellite to the GS}}, \quad \forall n \in \mathcal{N}, u \in \mathcal{U}_n. \end{aligned} \quad (3)$$

Stability-Aware SFC Orchestration and Associations

STARS at a glance

- **Stability-regularized objective:** penalize changes across time windows in α (user-sat), ξ (sat-GS), β (SFC).
- **Temporal decoupling:** warm-start from last window + visibility-aware *repair*.
- **Hierarchical decoupling:**
 - 1 **Stage I:** optimize associations (α, ξ) (LP).
 - 2 **Stage II:** optimize SFC placement & resources (β, Γ, χ) (reduced MINLP).



Stability-Aware SFC Orchestration and Associations

Objective Function

$$\min_{\alpha, \xi, \beta} \Psi = w_{\alpha} \cdot \Delta_{\alpha} + w_{\xi} \cdot \Delta_{\xi} + w_{\beta} \cdot \Delta_{\beta}, \quad (4)$$

subject to: the defined constraints, where the terms Δ_{α} , Δ_{ξ} , and Δ_{β} measure the temporal deviations from the previous configuration at time window $(\tau - 1)$ and are given by

$$\Delta_{\alpha} = \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}_n} \sum_{s \in \mathcal{S}} |\alpha_{u,n}^s - \alpha_{u,n}^s(\tau - 1)|, \quad (5)$$

$$\Delta_{\xi} = \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}_n} \sum_{s \in \mathcal{S}} \sum_{g \in \mathcal{G}} |\xi_{u,n}^{s,g} - \xi_{u,n}^{s,g}(\tau - 1)|, \quad (6)$$

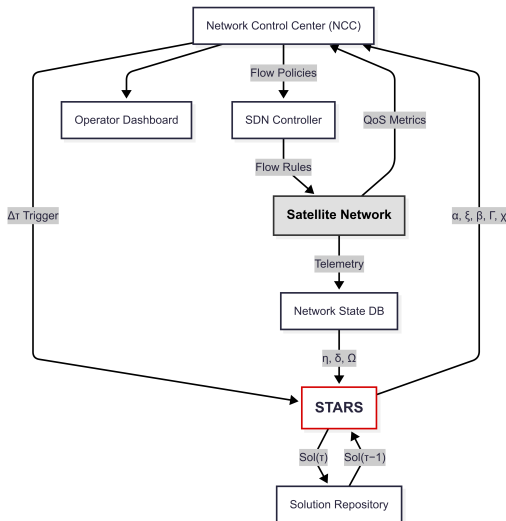
$$\Delta_{\beta} = \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}_n} \sum_{f \in \mathcal{F}_n} \sum_{i \in \mathcal{I}_f} \sum_{s \in \mathcal{S}} |\beta_{u,n,f}^{i,s} - \beta_{u,n,f}^{i,s}(\tau - 1)|. \quad (7)$$

- Normalize deviations to balance terms.
- Visibility filtering reduces search space.

Stability-Aware SFC Orchestration and Associations

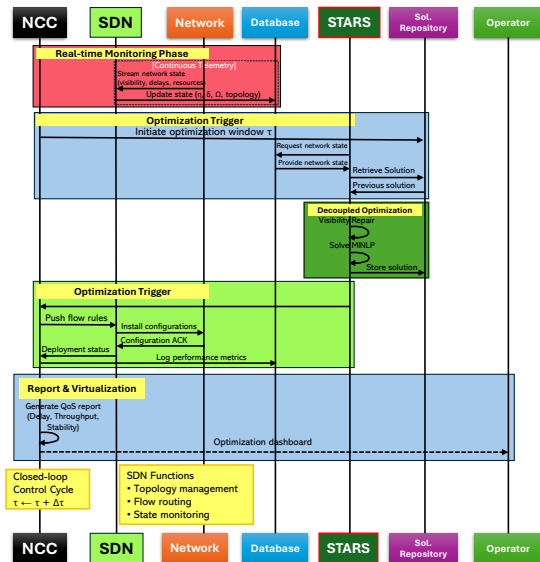
Closed-loop Orchestration

- 1 Telemetry: visibility, load, delay.
- 2 STARS computes the associations and SFC placements variables.
- 3 Provision via SDN/NCC, validate KPIs.



Stability-Aware SFC Orchestration and Associations

STARS Integration with NCC and SDN



Stability-Aware SFC Orchestration and Associations

Evaluation: Simulation setup [high level]

- Constellation: Walker-star, 60 sats, 4 planes, 500 km, 88° .
- Users: 30 (6 slices), two GSs.
- SFCs: 8 SF types, chains of length 2–4.
- Budgets: delay 35–150 ms, throughput 2–20 Mbps.
- Windowing: 14 windows, $\Delta\tau = 60$ s.
- Implementation: Julia + JuMP, Gurobi 11.0.

Stability-Aware SFC Orchestration and Associations

Evaluation: Compared approaches

- **MINLP** (monolithic; per-window; no stability).
- **MINLP-TD** (temporal penalties only).
- **MINLP-HD** (two-stage (hierarchical) only).
- **STARS** (temporal + hierarchical).

Stability-Aware SFC Orchestration and Associations

Results: Stability Improvements [selected plots]

Association churn

- STARS: ↓ ~ 54% *handovers*.
- HD only: ↓ ~ 18%.

SFC migrations

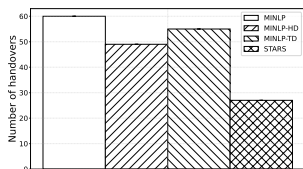
- STARS: ↓ ~ 33%.

Satellite-GS re-association

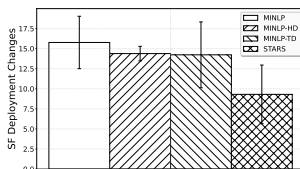
- STARS: ↓ ~ 59%.
- TD only: modest gains.

Runtime

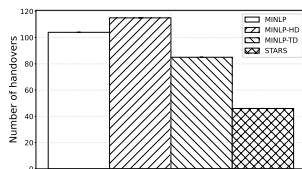
- STARS: ↓ ~ 57% vs. monolithic.



Association changes



SF deployment changes

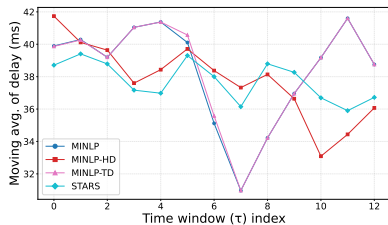
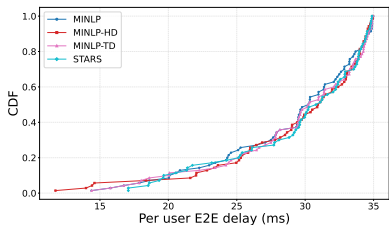


Satellite-GS changes

Stability-Aware SFC Orchestration and Associations

Results: Delay & Utilization [selected plots]

- **End-to-end delay:** *no degradation* vs. baselines and stability improvement (moving average $\sim 9\%$ lower).



- **CPU utilization:** better packing on sats ($\sim +10\%$ avg) while meeting QoS.

Stability-Aware SFC Orchestration and Associations

Key Takeaways and Future Work

- **Stability matters** in LEO: avoid unnecessary reconfigs.
- **STARS** couples stability penalties with temporal + hierarchical decoupling.
- **Results:** large reductions in churn and runtime, *no* delay penalty.
- **Future Work:** explore ML for **predictive**, ephemeris-aware orchestration.



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Risk-aware SFC Allocation

Proposed Solution

- The focus is on the optimal allocation of security functions (SFs) of different SFCs across a constellation of LEO satellites to support diverse slices.
- The proposed solution meets several key criteria:
 - Minimizing the end-to-end propagation delay .
 - Minimizing the number of SFs deployed across satellites
 - Minimizing the risk of sharing highly sensitive security functions between slices.
- This creates a trade-off:
 - Full sharing increases resource utilization but is also a high-security risk.
 - Full isolation minimizes risk but underutilizes resources.
- The shareability decision is established based on the risk assessment at the security function level.

Risk-aware Slicing-based Security Functions Allocation in LEO Satellite Networks

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Abstract—The integration of low Earth orbit (LEO) satellite communication into 6G networks promises a transformative impact on global connectivity by expanding coverage to remote regions and enhancing service reliability. However, this new infrastructure also introduces significant security challenges due to its expansive attack surface. To address this concern, we propose a dynamic security functions allocation (SFA) model that optimizes the allocation of security functions (SFs) across satellites while considering computational resource limitations, dynamic topology changes, and the visibility constraints of satellite constellations. Our model leverages the flexibility of 6G network slicing (NS) to share non-critical SFs between slices, reducing resource overhead while maintaining essential security demands. To minimize the risk of sharing highly sensitive SFs between slices, our model employs a nonlinear penalty, which prioritizes minimizing risk by aggressively penalizing high-risk

are conceptualized as virtual network functions (VNFs), are dynamically managed through software defined network (SDN) to ensure they meet the unique security requirements of each slice [8]. Previous studies have highlighted several key SFs such as encryption, decryption, firewalls, virtual private network (VPN), deep packet inspection (DPI), intrusion detection system (IDS), intrusion prevention system (IPS), security information and event management (SIEM) that should be deployed onboard satellites to provide comprehensive protection [9]. These virtual SFs are critical in enhancing network resilience by detecting and mitigating threats directly at the satellite level, reducing the risk of malicious activities spreading through the core network [10].

Risk-aware SFC Allocation

System Model: Dynamic Risk Assessment and Sharing Penalty

- Quantifies the potential impact of a security breach if a particular SF is shared between different slices.
- $P_{f,t}^{n,\acute{n}}$: Probability of a breach when SF f of type t is shared between slices n and $\acute{n}$.
- $I_{f,t}^{n,\acute{n}}$: Impact of a breach on the system.
- The risk $R_{f,t}^{n,\acute{n}}$ is calculated as:

$$R_{f,t}^{n,\acute{n}} = P_{f,t}^{n,\acute{n}} \times I_{f,t}^{n,\acute{n}} \quad (8)$$

- The penalty is computed as follows:

$$R = \sum_{\substack{n,\acute{n} \in \mathcal{N} \\ n \neq \acute{n}}} \sum_{\substack{f \in \mathcal{F}_n \\ \cap \mathcal{F}_{\acute{n}}}} \sum_{u \in \mathcal{U}_n} \sum_{\acute{u} \in \mathcal{U}_{\acute{n}}} \sum_{i \in I_f} \sum_{s \in S} \kappa \cdot (R_f^{n,\acute{n}})^2 \cdot \beta_{u,n,f}^{i,s} \cdot \beta_{\acute{u},\acute{n},f}^{i,s},$$

where, κ is a penalty coefficient that determines the weight applied to the squared risk values.

Risk-aware SFC Allocation

Problem Formulation: MINLP-based Optimization Approach

- Objective: Minimizing total traffic delay, the number of SFs installed, and the risk associated with the shared SFs across slices.

$$\min_{\substack{\beta, \Gamma \\ \alpha, \xi}} \left\{ \omega_d \cdot \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}_n} \delta_u^n + \omega_c \cdot \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} \gamma_{t,i}^s + \omega_r \cdot R \right\} \quad (9)$$

s.t.

$$\beta_{u,n,f}^{i,s}, \alpha_{u,n}^s, \xi_{u,n}^{s,g} \in \{0, 1\}, \quad \gamma_{t,i}^s, \Gamma_{u,n,f}^{i,s} \geq 0, \quad (10)$$

$$\sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{I}_f} \beta_{u,n,f}^{i,s} = 1 \quad \forall n \in \mathcal{N}, u \in \mathcal{U}_n, f \in \mathcal{F}_n, \quad (11)$$

$$\sum_{s \in \mathcal{S}} \alpha_{u,n}^s = 1 \quad \forall n \in \mathcal{N}, u \in \mathcal{U}_n, \quad (12)$$

$$\alpha_{u,n}^s \leq \eta_{u,n}^s \quad \forall n \in \mathcal{N}, u \in \mathcal{U}_n, s \in \mathcal{S}, \quad (13)$$

$$\sum_{s \in \mathcal{S}} \xi_{u,n}^s = 1 \quad \forall u \in \mathcal{U}_n, n \in \mathcal{N},$$

$$\xi_{u,n}^s \leq \eta_g^s \quad \forall n \in \mathcal{N}, u \in \mathcal{U}_n, g \in \mathcal{G}, s \in \mathcal{S},$$



Risk-aware SFC Allocation

Key Takeaways

- Prioritizing delay results in the least propagation delay.
- When capacity is prioritized, sharing is maximized, leading to a higher risk.
- When risk minimization is prioritized, sharing is limited, and only low-risk functions are shared between users.
- A balanced approach allows some sharing while limiting high-risk functions, providing flexibility to meet different objectives.

Risk-aware SFC Allocation (The Extension)

Risk Assessment and Sharing Penalty

- The proposed risk assessment framework quantifies the security risk of sharing a function as

$$R_{f,i,s}^{n,\hat{n}} = R_f \cdot E_{f,i,s}^{n,\hat{n}}, \quad (16)$$

- Function vulnerability level, $R_f = P_f \cdot M_f$,
 - $P_f \in [0, 1]$ is the probability of compromise for security function type f
 - $M_f \in [0, 10]$ is the impact magnitude of the breach on the system.
- Cross-slice exposure metric, $E_{f,i,s}^{n,\hat{n}}$.

$$E_{f,i,s}^{n,\hat{n}} = \underbrace{\max(C_n, C_{\hat{n}})}_{\text{criticality amplification}} \cdot \underbrace{\left(\mathcal{U}_n^{f,i,s} \cdot C_n \cdot \mathcal{U}_{\hat{n}}^{f,i,s} \cdot C_{\hat{n}} \right)}_{\text{weighted user exposure}}. \quad (17)$$

- The model minimizes the total risk

$$\mathcal{R}_{\text{total}} = \sum_{\substack{n, \hat{n} \in \mathcal{N} \\ n \neq \hat{n}}} \sum_{f \in \mathcal{F}_n \cap \mathcal{F}_{\hat{n}}} \sum_{i \in \mathcal{I}_f} \sum_{s \in \mathcal{S}} R_{f,i,s}^{n,\hat{n}}.$$



Risk-aware SFC Allocation (The Extension)

Controlled Instance Splitting

- To ensure that, *by default*, all users of the same slice n share the same (f, i, s) instance, while still allowing extra instances if capacity, throughput, or delay constraints necessitate, we introduce:

$$X_{n,f}^{i,s} \in \{0, 1\} \quad \text{and} \quad \text{Slack}_n^f \in \mathbb{Z}_{\geq 0},$$

where

$$\beta_{u,n,f}^{i,s} \leq X_{n,f}^{i,s}, \quad (19)$$

$$\sum_{i \in \mathcal{I}_f} \sum_{s \in \mathcal{S}} X_{n,f}^{i,s} \leq 1 + \text{Slack}_n^f. \quad (20)$$

Here, $X_{n,f}^{i,s} = 1$ if slice n uses instance (f, i) on satellite s , and Slack_n^f counts any extra instances beyond the first shared instance.

Risk-aware SFC Allocation (The Extension)

Objective Function

- The objective function in our model, shown in (21), aims to minimize the risk associated with the shared SF across slices.

$$\min_{\beta, \Gamma, \alpha} \left\{ \mathcal{R}_{\text{total}} + \sum_{n \in \mathcal{N}} \sum_{f \in \mathcal{F}_n} \text{Slack}_n^f \right\}, \quad (21)$$

Subject to the defined constraints.

Implementation

- To tackle the complexity of solving this problem, we implemented the model using the **Simulated Annealing Metaheuristic**.

Security Function Chain Provisioning in LEO Satellite Networks: A Risk-Driven Optimization

Authors' Draft: Work in Progress November 15, 2025

Abstract—The integration of low Earth orbit (LEO) satellite communication into 6G networks has the potential to revolutionize global connectivity by extending coverage to underserved remote areas and enhancing service reliability. However, this innovation introduces substantial security challenges due to the expanded attack surface inherent in satellite-based systems. To address these challenges, we propose a dynamic security functions allocation (SFA) model designed to optimize the placement of security functions (SFs) across satellites while accounting for constraints such as limited computational resources, dynamic topology changes, and visibility limitations within satellite constellations. Our model harnesses the flexibility of 6G network slicing (NS) to share non-critical SFs between slices, thereby reducing resource consumption without compromising essential security requirements. To mitigate risks associated with sharing highly sensitive SFs, the model incorporates a nonlinear penalty mechanism that prioritizes minimizing risks by heavily penalizing the sharing of high-risk SFs. This dynamic risk management framework evaluates both the likelihood and impact of potential security breaches, ensuring SFs are shared only under acceptable

the needs of each application, reducing resource wastage and optimizing the system's overall performance [9]. To this end, 3rd generation partnership project (3GPP) proposed NS as a key enabler for software-defined 6G networks [10].

However, as the scale and complexity of 6G satellite networks increase, the attack surface grows, demanding robust and innovative security mechanisms to protect sensitive communications and maintain network integrity [11]. Recent incidents have highlighted the vulnerabilities in satellite communication systems and network slicing, underscoring the importance of robust security measures. In early 2022, a cyberattack targeted Viasat's KA-SAT satellite broadband service, causing widespread internet outages across Ukraine and parts of Europe. Researchers also discovered that SpaceX's Starlink satellite terminals were susceptible to hacking through a custom modchip, which allowed attackers to execute arbitrary code. Additionally, persistent disturbances

Talk Roadmap

- 1 Introduction and Research Problems
- 2 Visibility-Aware User Association and Resource Allocation
- 3 Stability-Aware Security Function Chain (SFC) Orchestration and Associations
- 4 Risk-aware SFC Allocation
- 5 Conclusion**
- 6 Questions

Conclusion

- We addressed fair and stable user association in dynamic satellite networks.
- We tackle the joint problem of user–satellite association, satellite–ground station (GS) association, and SFC placement in a dynamic LEO network.
- We proposed risk-aware function allocation for satellite networks that balances efficiency and security.

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Questions

Thank you for your attention!

